

iRevolution: A Data-Driven Exploration of Apple's iPhone Impact in India Using Tableau

1. Introduction

Project Title: iRevolution: A Data-Driven Exploration of Apple's iPhone Impact in India Using Tableau

Team Members:

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Team Size: 5

Team Leader: Kanta Sruthi

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2. Project Overview

Purpose:

The main purpose of this project is to analyze product-related data and generate actionable insights. It helps in understanding sales trends, price distribution, and product popularity. The project also demonstrates how business intelligence tools can be integrated with web technologies. It enhances practical knowledge of full-stack development concepts. Additionally, it improves skills in data visualization and deployment.

Features:

1. Interactive Tableau dashboards
2. Product sales and price analysis
3. Embedded dashboard in Flask web app
4. Responsive HTML interface
5. Localhost web deployment

3. Architecture

Frontend:

The front-end consists of HTML templates rendered using Flask. Bootstrap-based styling is used to improve the visual appearance of the webpage. The embedded Tableau dashboard appears inside the HTML page using iframe/object tags. The interface is clean and simple for better usability. Users can access the dashboard through a web browser

Backend:

The backend is developed using Flask, a lightweight Python web framework. Flask handles routing and rendering of HTML templates. The main application file (app.py) defines the routes and runs the local server. The server runs on localhost (127.0.0.1:5000). It ensures proper loading of the embedded Tableau dashboard.

Database:

The project uses a file-based dataset (Excel/CSV) as the primary data source. The dataset is connected directly to Tableau Desktop for data visualization and analysis. No external database server such as MySQL or SQL Server is used in this project. The data is cleaned and processed within Tableau before creating dashboards and stories. After publishing to Tableau Public, the processed data is rendered through the embedded visualization in the Flask web application. This lightweight approach ensures simplicity, easy deployment, and efficient performance for academic use.

4.Setup Instructions

Prerequisites:

- 1.Python installed
- 2.Flask library installed
- 3.Tableau Public/Desktop
- 4.Basic HTML & Python knowledge
- 5.Internet connection

Installation / Setup Steps:

- 1.Install Python
- 2.Install Flask using pip install flask
- 3.Create project folder
- 4.Add app.py and index.html
- 5.Paste Tableau embed code

5.Folder Structure

- 1.app.py (main backend file)
- 2.Templates/ folder
- 3.index.html inside templates
- 4.static/ folder (CSS/JS if needed)

6.Running the Application

1. Open project in VS Code
2. Open terminal
3. Run python app.py
4. Open browser
5. Visit <http://127.0.0.1:5000>

7.PI Documentation

The project uses Flask routing as a simple API mechanism. The main route "/" renders the index.html page. No external APIs are integrated in this project. Tableau embed JavaScript API is used for rendering visualizations. The application works as a basic web service hosting the dashboard

8.Authentication

Currently, the application does not implement user authentication. The dashboard is publicly accessible through Tableau Public. Security is minimal since it runs on localhost. For production deployment, authentication mechanisms can be added. Future versions can include login systems for restricted access.

9. User Interface

The user interface is simple and responsive. It includes a structured layout with headers and embedded visualizations. Users can interact with filters and charts within the Tableau dashboard. The interface ensures smooth navigation and readability. The design enhances user experience and accessibility.

10. Testing

1. Start Flask server
2. Check dashboard loads correctly
3. Test filters and interactions
4. Verify no console errors
5. Ensure internet connectivity

11. Known Issues

1. Requires internet connection
2. Works only when server is running
3. No user authentication
4. Performance depends on dataset size

12. Future Enhancements

1. Add login authentication
2. Connect real-time database
3. Deploy to cloud server
4. Improve UI design
5. Add advanced analytics